Linear regression

- Regression = supervised prediction of *real* labels
  - data: features $X^t \in \mathbb{R}^d$, labels $Y^t \in \mathbb{R}$, $t = 1 : T$
- *Linear* regression = predict a linear function of features
  - model: predict $\hat{Y}^t = w \cdot X^t = \sum_i w_i X_i^t$ or $w \cdot X^t + b$
  - can omit b if we have a constant feature

Model complexity

- Model complexity (for model selection):
  - how many features/weights
  - how many *nonzero* weights
  - *size* of weights (there are more effectively-distinct weight vectors in $\{w \mid \|w\| \leq 100\}$ than in $\{w \mid \|w\| \leq 1\}$)

Size and count are related

Linear regression

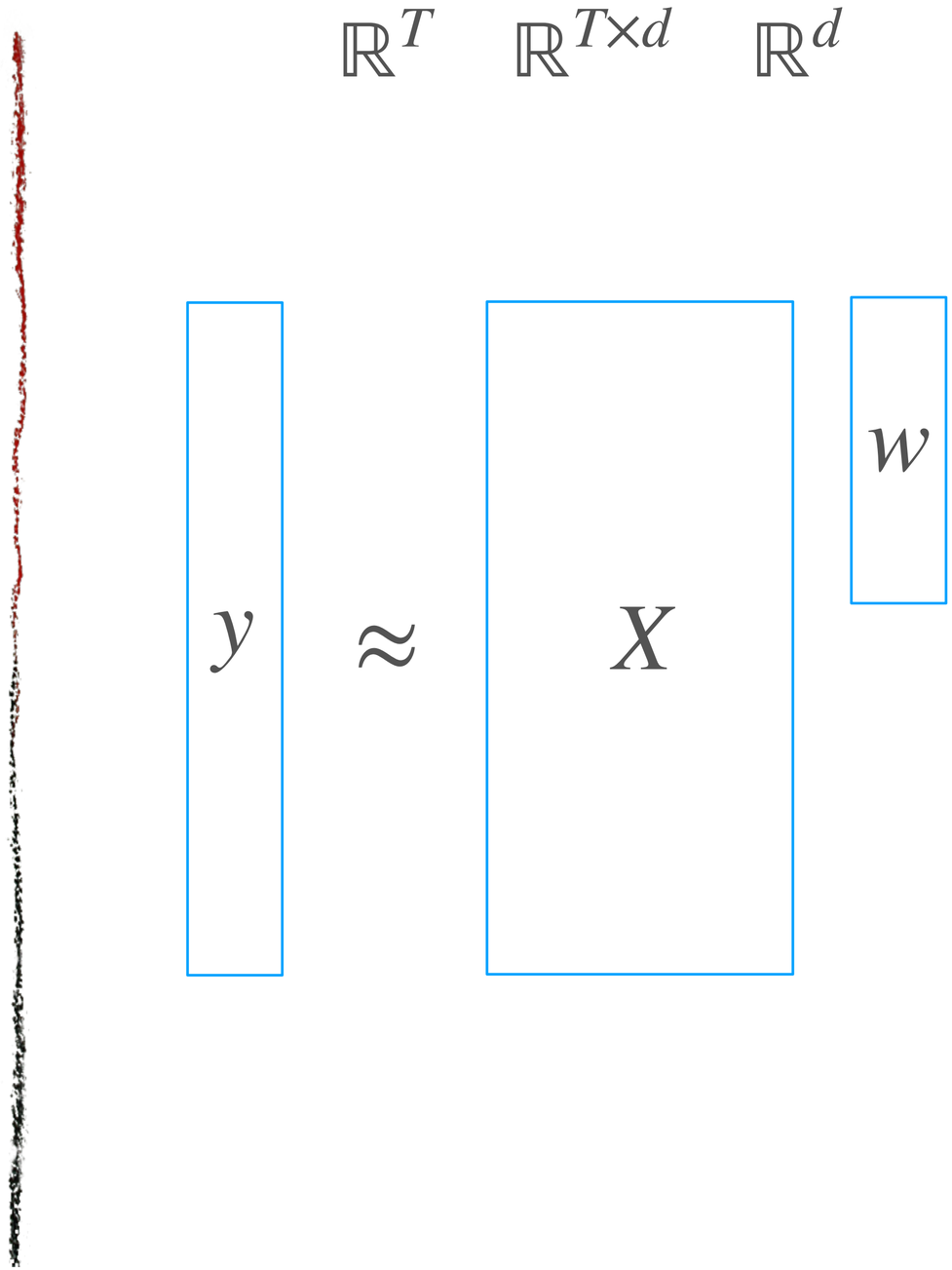
- Objective:
  - minimize sum of squared errors: $\frac{1}{2}\sum_{t=1}^{T}(Y^t - \hat{Y}^t)^2$
  - plus optional regularizer: e.g., L_2, L_1, elastic net
  - $\frac{\lambda}{2}||w||^2$ called *ridge regression*
- E.g., minimize

$$L(w) = \frac{1}{2}\sum_{t=1}^{T}(Y^t - \hat{Y}^t)^2 + \frac{\lambda}{2}||w||_1^2$$

Solving it

- To find best w, several options:
  - minimize symbolically by setting $\nabla_w L$ to 0
    - sub-choices: QR, SVD, Cholesky
  - iterative optimizer
    - sub-choices: stochastic vs. full gradient
    - 1st-order like SGD or Adam
    - 2nd-order like Newton
    - in-between like L-BFGS
    - tools like momentum, rescaling
  - choice of covariance form or kernel form

Schematic

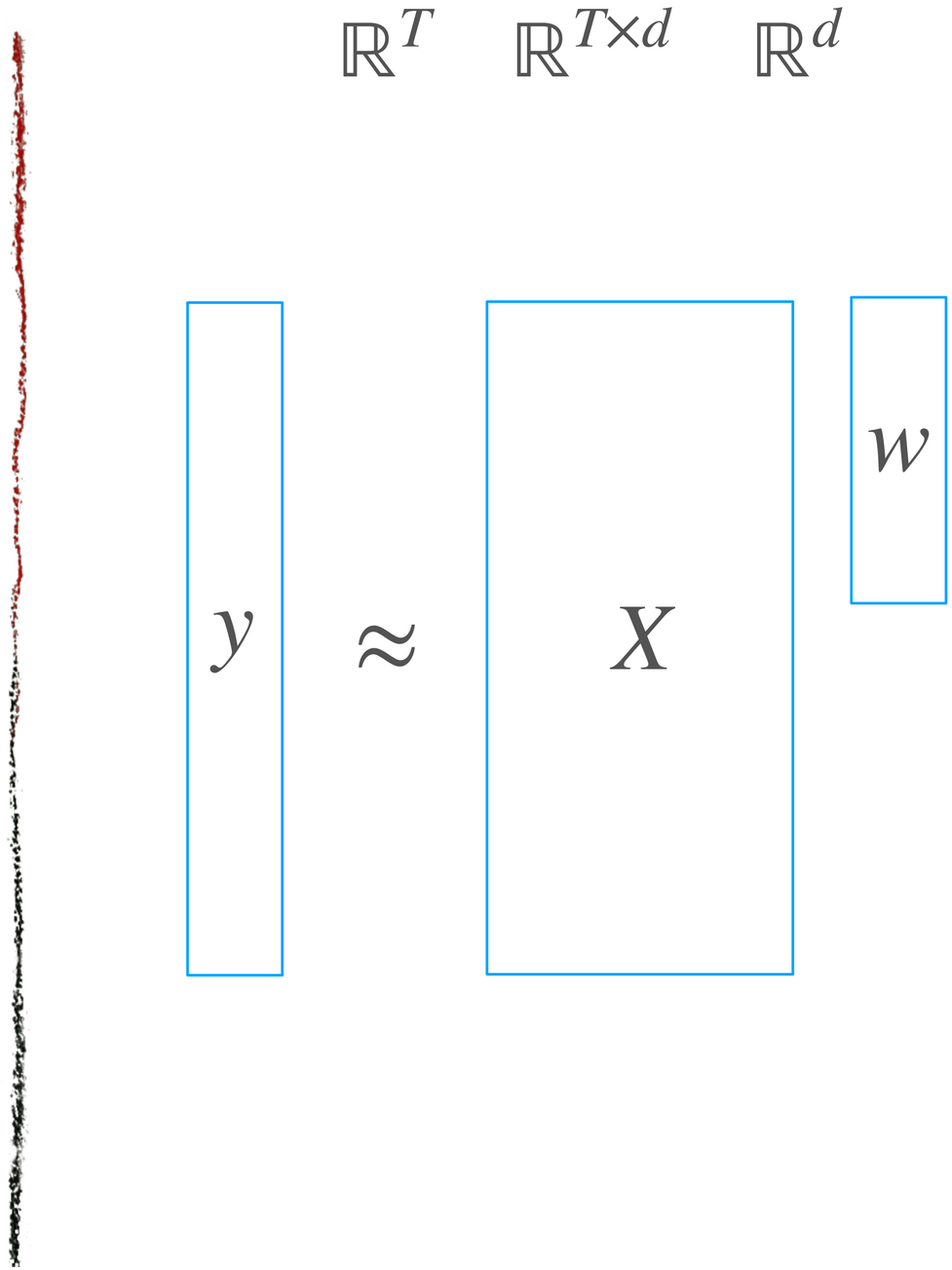


features labels

Family History	Resting Blood Pressure	Cholesterol	Cost (k$)
Yes	Low	Normal	0.1
No	Medium	Normal	0.2
No	Low	Abnormal	10.3
Yes	Medium	Normal	8.7
Yes	High	Abnormal	13.1
No	Low	Normal	0.1
No	Medium	Normal	0.4
Yes	Medium	Abnormal	9.6

- data matrix X, label vector Y; rows are X^t, Y^t

Schematic



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Schematic

$$\mathbb{R}^T \quad \mathbb{R}^{T\times d} \quad \mathbb{R}^d$$

$$y \approx X \; w$$

$$L(w) = \|y - Xw\|_F^2 \quad [\,+ R(w)\,]$$

features | labels

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MOO!

- Previous slides illustrate *model, objective, optimizer* for linear regression
- Often three good questions to ask when learning a new ML method — maybe not complete, but a good place to start

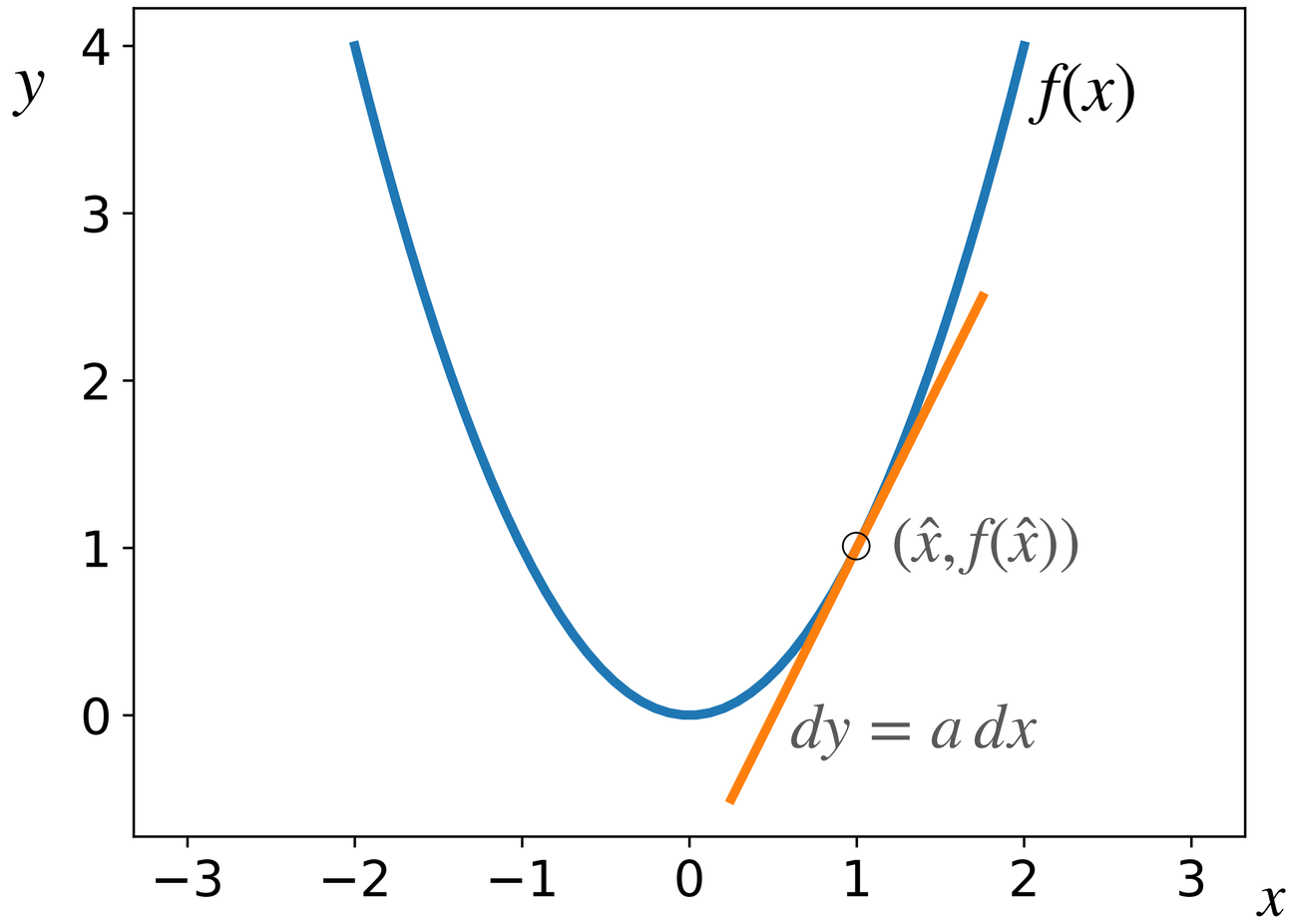
Gradients

- For any of the above, need gradients
  - $L(w) = \frac{\lambda}{2}\|w\|^2 + \frac{1}{2}\sum_{t=1}^{T}(Y^t - w \cdot X^t)^2$
  - $\nabla L(w) =$

Derivatives everywhere

- The way we derive many ML methods:
  - start from objective to minimize (logistic loss or sum of squared errors, plus optional regularizer)
  - take a few derivatives ($0 = d\,\text{objective} / dw$)
  - find optimum (e.g., by SGD)
  - done!
- Will continue to be a common pattern
- Review of intuition and tools for high-d derivatives, so we can do it fast

Derivatives notation

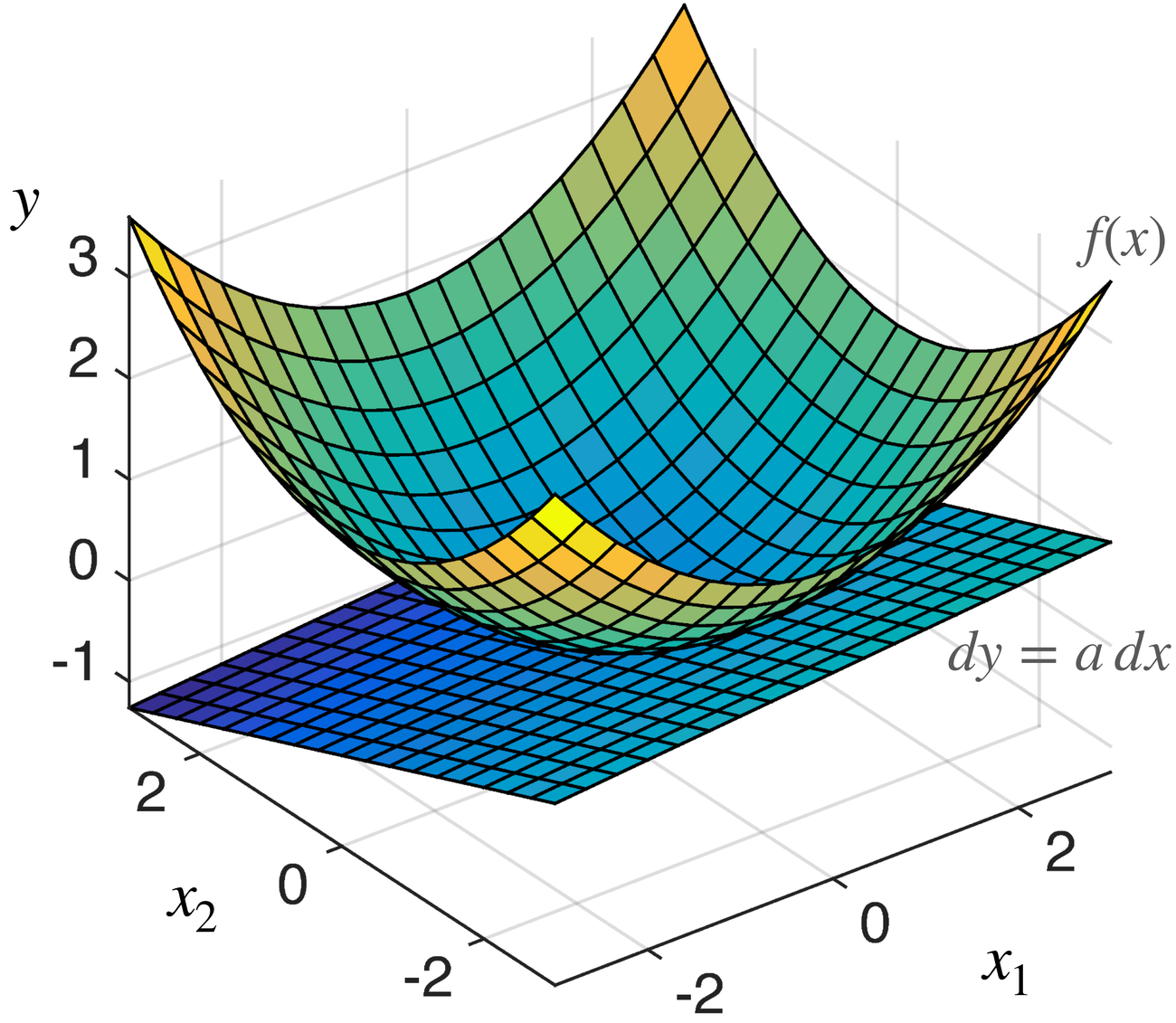


$m = n = 1$

$dy = y - f(\hat{x})$
$dx = x - \hat{x}$

- Function $y = f(x)$
  - $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, $f \in \mathbb{R}^n \to \mathbb{R}^m$
- Derivative is a *linear fn* that *locally* approximates f near $\hat{x}$
  - $dy = a\,dx$, where $a \in \mathbb{R}$ is the derivative

Higher dimensions

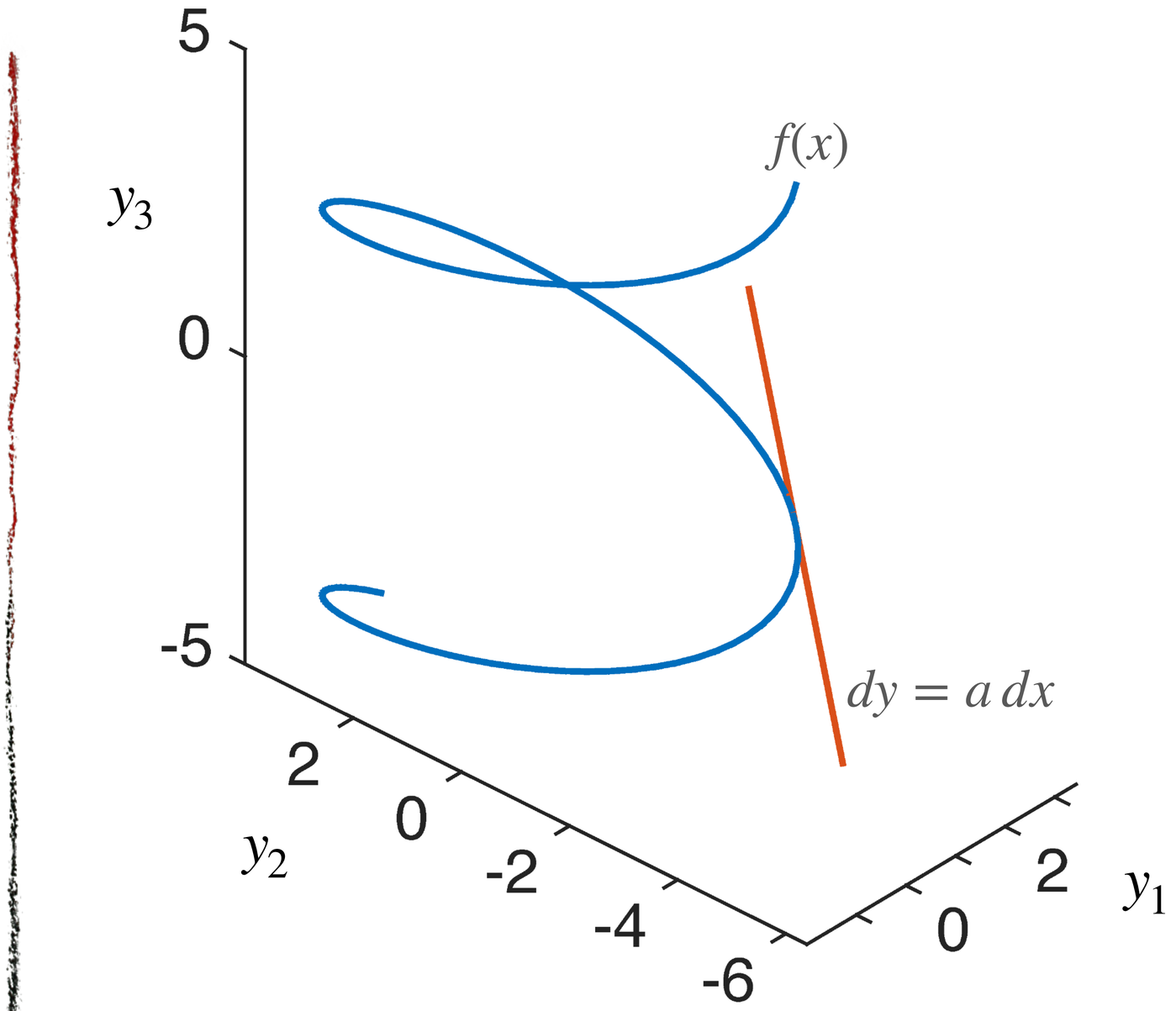


$\mathbb{R}^n \to \mathbb{R}^m$

$m = 1, n = 2$

Function $y = f(x)$ $\quad dy = a\,dx \quad a \in \mathbb{R}^{1\times 2}$

Higher dimensions



$\mathbb{R}^n \to \mathbb{R}^m$

$m = 3, n = 1$

Function $y = f(x)$ $\quad dy = a\,dx \quad a \in \mathbb{R}^{3\times 1}$

Notation for linear functions

output shape (derivative of) \ input shape (with respect to)	Scalar	Vector	Matrix
Scalar	$ds = a\,dt$	$ds = u^T dv$	$ds = \mathrm{tr}(M^T dN)$
Vector	$du = v\,ds$	$du = M\,dv$	$\times$
Matrix	$dM = N\,ds$	$\times$	$\times$

- a, s, t: scalars u, v: column vectors M, N: matrices
- note: $ds = a\,dt$ is a synonym for $a = \frac{ds}{dt}$, etc.
- $\times$ means no common notation; see torch.einsum
- derivative = coefficient of ds, dt, dv, dN on RHS

Notation for linear functions

output shape (derivative of) \ input shape (with respect to)	Scalar	Vector	Matrix
Scalar	$ds = a\,dt$	gradient	$ds = \mathrm{tr}(M^T dN)$
Vector	velocity	Jacobian	$\times$
Matrix	$dM = N\,ds$	$\times$	$\times$

- a, s, t: scalars u, v: column vectors M, N: matrices
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Tools for derivatives: trace

- Trace of a matrix: tr$(A) = \sum_i A_{ii}$
  - scalar-valued linear function of a square matrix
- Lets us write “matrix dot product”
  - old way (subscripts): $\sum_{ij} A_{ij} B_{ij}$
  - easier way: tr$(A^\top B)$

$[A^\top B]_{ik} =$

Trace identities

- scalar = scalar$^\top$ = tr(scalar)
- Trace rotation
  - tr(ABC) = tr(CAB) when A, B, C have compatible dimensions
- Trace-Hadamard
  - $A \circ B =$ componentwise product, $[A \circ B]_{ij} = A_{ij} B_{ij}$
  - tr $A^\top (B \circ C) =$ tr $(A \circ B)^\top C$

Tools for derivatives: linearity, chain, product

- Linearity:
  - $d(ax+b) = a\,dx + b, \quad d(AX+B) = A\,dX + B, \ldots$
- Chain rule for $f(g(x))$
  - old notation: $\frac{df}{dx} = \frac{df}{dg}\frac{dg}{dx}$ ("cancel the dg")
  - new: if $df = A\,dg$, $dg = B\,dx$ then $df =$
- Product rule for $f(x)\,g(x)$
  - $d(fg) = df\,g + f\,dg$
  - works for any kind of product (matrix product, vector cross product, elementwise (Hadamard) product, vector convolution, Khatri-Rao product …)
  - unlike scalar case, order can matter (products are not necessarily commutative)

Partial vs. total derivatives

- In a function of several variables, partial derivative imagines changing one of them, holding others fixed
  - e.g., $\frac{\partial}{\partial x}(x+y)^2 =$ holding y fixed
- New notation: any variable that appears with d in RHS is changing, others are fixed
  - e.g., $d(x+y+z)^2 =$ holding z fixed
- If variables appear together, law of total derivatives

$$\begin{aligned} df(u(x), v(x), w(x)) = & f^{(1)}(u,v,w)\, u'(x)\, dx \\ & + f^{(2)}(u,v,w)\, v'(x)\, dx \\ & + f^{(3)}(u,v,w)\, w'(x)\, dx \end{aligned}$$

  - e.g., $d(u(x)+v(x)+w(x))^2 =$

Gradient practice: Linear regression in one slide

$$L = \|Y - Xw\|^2 = (Y - Xw)^\top (Y - Xw)$$

$$L = Y^\top Y - 2Y^\top Xw + w^\top X^\top Xw$$